LECTURE 10: BERTRAND COMPETITION

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WEEK 4

LAST LECTURE WE LEARNED ABOUT COURNOT COMPETITION

- Cournot competition occurs when firms compete based on quantity
- The other natural way to compete is by price competition (**I'm about to release my catalogue, at what price should I list my products?**)
- We call this Bertrand competition
- We'll see we actually get very different outcomes compared to Cournot (quantity) models

BUT FIRST, WHO WANTS TO WIN $5?

- I want to auction off a $5 note
- Each player takes turns making an offer to buy my money
- Offers need to be whole dollars $1, $2, etc
- Winning bid earns $\$5 - \text{Highest bid}$
- Loser gets $0

BUT FIRST, WHO WANTS TO WIN $5?

- Assuming this went well, we should have seen bids go up to $5, eliminating any margin
- Any player with a losing bid should have an incentive to bid $1 more than the other player up to $5
- Let's think about the best response function here:

$$BR_i = \begin{cases} \text{bid}_j + 1 & \text{bid}_j < \$5 \\ \$5 & \text{Otherwise} \end{cases}$$

- The best response function shows that each player would always want to undercut the other player

THIS IS ESSENTIALLY HOW BERTRAND COMPETITION WORKS

- In Bertrand modelling with identical products, the cheapest price takes the whole market, and it's split when prices are the same:

$$q_1 = \begin{cases} 0 & p_1 > p_2 \quad \text{Firm 1 gets nothing if they price too high} \\ Q_D/2 & p_1 = p_2 \quad \text{The market is split} \\ Q_D & p_1 < p_2 \quad \text{Firm 1 gets the whole market if they undercut Firm 2} \end{cases}$$

- We could think of this as an auction, but instead of the auctioneer driving the price higher, competition drives it lower

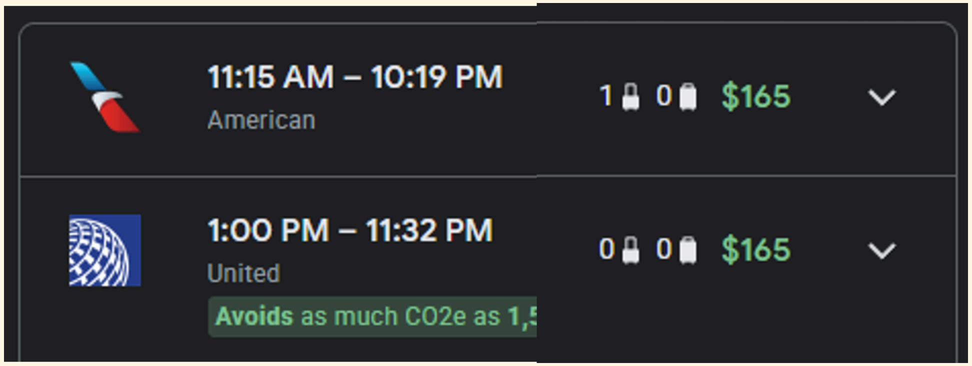
THIS IS ESSENTIALLY HOW BERTRAND COMPETITION WORKS

- Take two firms, both of which have marginal costs of $5
- What are each firm's best responses here if they compete on price?
- If Firm 1 prices at $7, Firm 2 could undercut them at $6 and take the whole market
- If Firm 2 is pricing at $6, Firm 1 could undercut them at $5, its marginal cost
- Firm 2 would rather split the market so also offers at MC

In Bertrand with identical goods, the price always falls to the marginal cost - Firms always have the incentive to compete with one another

WAIT, ISN'T THIS DIFFERENT TO WHAT WAS PREDICTED BY COURNOT?

- In Cournot (quantity) games with identical products, we learnt there'll always be a mark up above marginal cost
- We just showed that our Bertrand (price) games with identical products, price will fall to marginal cost
- What gives?
- My take: Each model is better at describing different markets better than others
- Bertrand describe some markets like gas stations, car insurance, or particular airline routes better than Cournot - Markets where you migth expect margins to get squeezed over time



BEFORE WE GO MUCH FURTHER - ANOTHER $5 AUCTION

But now the rules are slightly changed

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- Each player takes turns making an offer to buy my money
- Offers need to be whole dollars $1, $2, etc
- Winning bid earns $\$5 - \text{Highest bid}$
- **Loser still pays their losing bid!**

BEFORE WE GO MUCH FURTHER - ANOTHER $5 AUCTION

- Assuming that went well, we should have gotten a pretty weird outcome
- We should have gotten to a point where its a better strategy to bid more than $5
- Actually only a Nash Equilibrium strategy in mixed strategies (randomisation) where people should expect to lose money most of the time
- All this is to say, slightly changing the rule in competition/auctions/games can create massive differences in outcomes

LIKE COURNOT, WE CAN THINK ABOUT BERTRAND COMPETITION WITH DIFFERENTIATED GOODS

Price won't fall to marginal cost where we have differentiated products

Let's see how Coke and Pepsi might compete on price:

- The demand for coke is: $q_C = 58 - 4P_C + 2P_P$
- The demand for pepsi is: $q_P = 63.2 - 4P_P + 1.6P_C$

Let total cost for both coke and pepsi both be:

$$TC_C = 5q_C$$
$$TC_P = 5q_P$$

LIKE COURNOT, WE CAN THINK ABOUT BERTRAND COMPETITION WITH DIFFERENTIATED GOODS

Very similar approach to Cournot (quantity) modelling, but we frame everything in terms of choosing prices!

1 - Setup our profit maximisation problem for each firm - needs to consider the prices of the other firms! Each firm has their own demand curve

2 - Find the best response functions - at what price should each firm charge as a function of other firm's price

3 - Find where the best response functions coincide/intersect - this is a Nash-Bertrand equilibrium

LET'S FIND COKE'S BEST RESPONSE FUNCTION

Coke wants to choose its price to maximise its profit (But not a monopoly!)

$$
\begin{aligned}
\max_{P_C} \pi_C &= \max_{P_C} P_C q_C - 5q_C \\
&= \max_{P_C} P_C(58 - 4P_C + 2P_P) - 5(58 - 4P_C + 2P_P) \\
&= \max_{P_C} P_C(58 - 4P_C + 2P_P) - 5(58 - 4P_C + 2P_P) \\
&= \max_{P_C} 58P_C - 4P_C^2 + 2P_P P_C - 290 + 20P_C - 10P_P \\
&= \max_{P_C} 78P_C - 4P_C^2 + 2P_P P_C - 290 - 10P_P
\end{aligned}
$$

We find the first order condition:

$$
\frac{d\pi_C}{dP_C} = 78 - 8P_C + 2P_P = 0
$$

$$
\begin{aligned}
8P_C &= 78 + 2P_P \\
P_C &= BR_C = 9.75 + 0.25P_P
\end{aligned}
$$

NOW LET'S FIND PEPSI'S BEST RESPONSE FUNCTION

Pepsi wants to choose its price to maximise its profit (But not a monopoly!)

$$\begin{aligned}
\max_{P_P} \pi_P &= \max_{P_P} P_P q_P - 5q_P \\
&= \max_{P_P} P_P(63.2 - 4P_P + 1.6P_C) - 5(63.2 - 4P_P + 1.6P_C) \\
&= \max_{P_P} 63.2P_P - 4P_P^2 + 1.6P_C P_P - 316 + 20P_P - 8P_C \\
&= \max_{P_P} 83.2P_P - 4P_P^2 + 1.6P_C P_P - 316 - 8P_C
\end{aligned}$$

We find the first order condition:

$$\frac{d\pi_P}{dP_P} = 83.2 - 8P_P + 1.6P_C = 0$$

$$\begin{aligned}
8P_P &= 83.2 + 1.6P_C \\
P_P &= BR_P = 10.4 + 0.2P_C
\end{aligned}$$

FINDING THE NASH-BERTRAND EQUILIBRIUM

We have both of our best response functions:

$$\begin{aligned} P_C &= BR_C = 9.75 + 0.25P_P \\ P_P &= BR_P = 10.4 + 0.2P_C \end{aligned}$$

Where they intersect is the Nash equilibrium:

$$\begin{aligned} P_C &= 9.75 + 0.25(10.4 + 0.2P_C) \\ P_C &= 9.75 + 2.6 + 0.05P_C \\ P_C &= 12.35 + 0.05P_C \\ 0.95P_C &= 12.35 \\ P_C &= 13 \end{aligned}$$

So then, $P_P = 10.4 + 0.2 \times 13 = 13$

Result: Both Coke and Pepsi charge $13 in equilibrium! Even though they have differentiated products and slightly different demand curves, the Nash equilibrium results in identical pricing.